Master of Data Analytics, University of Niagara Falls, Canada

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Report

Group 8:

Frigga Hannah Dawson Johnson - NF1021758

Gabriel Romero Gutierrez - NF 1012849

Lilian Ngozi Ugwu-NF1016558

Mihir Bhadreshkumar Parikh – NF1011072

Professor: Ph.D Payman Janbakhsh

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Abstract
An exploratory analysis of a loan application dataset comprising records with credit, financial, and demographic data is presented in this report. The objective is to identify key trends and patterns that may influence loan approval. Data preprocessing addressed missing values, corrected inconsistencies, encoded categorical variables, and applied log transformations to reduce skewness in income and loan amount distributions. Despite male predominance, preliminary results indicate that married applicants, graduates, and semi-urban residents had higher approval rates. There is also no indication of gender bias. Chi-square tests show that loan approval is significantly correlated with education level, property location, and credit history (for medium- and long-term loans). These insights suggest that certain features may have substantial predictive power. Model development and performance evaluation will be conducted in subsequent phases to build a robust and fair loan eligibility prediction system.

Problem Definition

Loan Approval Prediction

Problem Statement

The goal of this project is to create a machine learning-based system that uses loan applicants' credit, financial, and demographic data to forecast their eligibility. Indicators including applicant and co-applicant income, loan amount, loan length, credit history, employment status, and educational attainment, are among the historical loan application data included in the dataset.

The project will use this data to train models to categorize applicants as either eligible (approved) or not eligible (rejected) for loans, as well as investigate how different features affect loan approval choices.

Objectives:

- Construct a prediction model to categorize loan applicants as eligible or ineligible.
- Determine the main factors influencing loan approval decisions.
- Effectively handle and preprocess category and missing data.
- Use metrics such as recall, accuracy, and precision to assess the model's performance.
- Examine equity and possible prejudices among various demographic groupings.

Methodology:
1. Analysis of Data: Recognize missing values, examine category variables, and comprehend distributions.To determine approval rates, examine the goal balance (Loan_Status).
2. Preparing the data: Impute missing values. Use methods like OneHot or Label Encoding to encode categorical features (such as gender and education). If necessary, normalize or scale numerical data.

3. Model Building: Train different classifiers, such as Random Forest, Decision Trees, and Logistic Regression. Do cross-validation to make sure it's resilient.
4. Evaluation: Make use of the F1-score, recall, accuracy, precision, and confusion matrix. Analyze models and visualize performance.
5. Interpretation: Determine the significance of the extracted features.

Research questions and Hypotheses

Research Question 1: Does loan term category by credit history significantly affect loan approval rates?

- Null hypothesis (Ho): Credit history does not significantly influence the likelihood of loan approval.
- Alternative hypothesis (H1): Credit history significantly influences the likelihood of loan approval.

Research Question 2: Is there a difference in loan approval rate based on educational level?

- Null hypothesis (Ho): Educational level has no influence on loan approval
- Alternative Hypothesis (H1): There is a significant influence of educational level on loan approval.

Research Question 3: Does property location such as urban, rural and semiurban areas impact loan eligibility?

- Null hypothesis (H0): Property location does not influence loan approval.
- Alternative Hypothesis (H1): Property location significantly impacts loan approval.

Data Cleanning

The dataset which we used for the analysis consisted of 614 rows and 13 columns related to loan applications, containing both applicant information and financial details. Before proceeding with any analysis, a thorough data cleaning process was undertaken to ensure the dataset was accurate, consistent, and suitable for further exploration.

The initial assessment revealed missing values in several columns. These included categorical features such as Gender, Married, Dependents, and Self_Employed, as well as numerical fields like LoanAmount, Loan_Amount_Term, and Credit_History. To handle these missing values effectively, a strategy was adopted based on the nature of each variable. Categorical columns were filled using their mode, preserving the distribution of the data. For numerical columns, the median was used to avoid skewing the data, especially in the presence of outliers.

An inconsistency was identified in the Dependents column, where the value '3+' was present. To maintain numerical uniformity and facilitate further transformation, this value was replaced with '3', and the column was converted to a numeric data type.

Outlier detection was conducted using the Interquartile Range (IQR) method. This revealed that ApplicantIncome, CoapplicantIncome, and LoanAmount contained a noticeable percentage of outliers. Higher outlier ratios were also found in Loan_Amount_Term and Credit_History. However, instead of removing these rows—particularly from columns with high proportions of outliers—log transformations were applied to ApplicantIncome, CoapplicantIncome, and LoanAmount. This approach helped in reducing skewness and mitigating the influence of extreme values without compromising data volume.

To enhance the interpretability of the Loan_Amount_Term feature, a new categorical column named Loan_Term_Category was created. This column categorizes loan durations into three segments: Short Term (loans with terms up to 120 months), Mid Term (121 to 300 months), and Long Term (over 300 months). This derived feature simplifies the analysis of loan distribution over different durations.

The dataset is now structured, reliable, and ready for use in further stages of the project, analysis through visualization.

Data Understanding and Visualizations

The initial dataset analysis involved identifying the data type for each variable as follows:

Discrete Variables: *Loan_ID, Gender (Categorical), Married (Binary), Dependents (Categorical), Education (Categorical), Self_Employed (Binary), Loan_Amount_Term (Categorical), Loan_Amount_Term (Categorical), Credit_History (Categorical), and the objective variable Loan_Status (Binary)*

To understand the distribution and behavior of the categorical variables related to the objective variable, grouped bar charts were constructed to show the relationship between them. These charts clearly show how the proportion of approved and unapproved loans varies according to each category.

In addition, pie charts were created to visualize the overall distribution of each categorical variable in the dataset, regardless of loan status. These visualizations provide a quick and intuitive perspective on the balance or imbalance between categories and allow us to identify possible biases or dominance of certain classes that could influence subsequent predictive models.

Overall, the approved loans distribution (Figure 1) suggests that the dataset contains a greater number of records where loans were approved. This observation remains constant regardless of gender, with both men and women receiving more approvals than rejections.

The demographic breakdown shows patterns that match earlier statistical findings. Graduates (Figure 2) still have a much higher approval rate than non-graduates, who do not receive any approvals in this filtered sample. Married applicants are more favorably positioned.

Also, gender does not show any signs of bias, even with numerical imbalances in representation. Dependents continue to be a factor, although their impact is lessened in this smaller dataset.

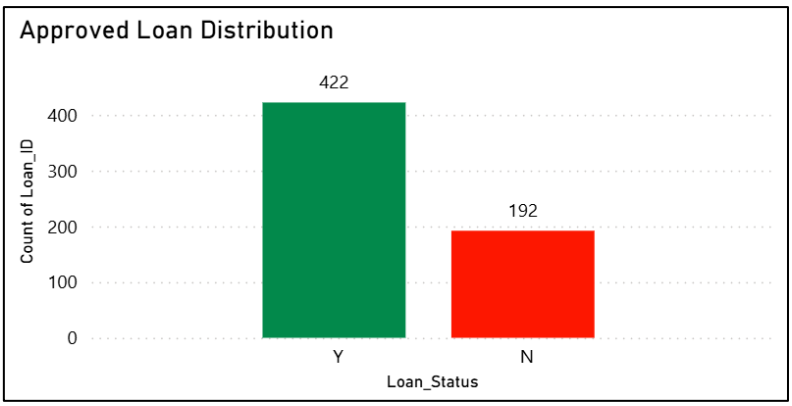


Figure 1. Approved loans dataset distribution

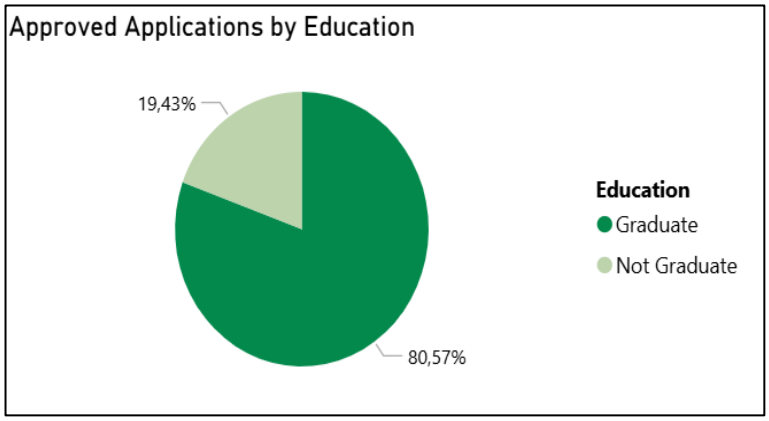


Figure 2. Approved Loans by Education

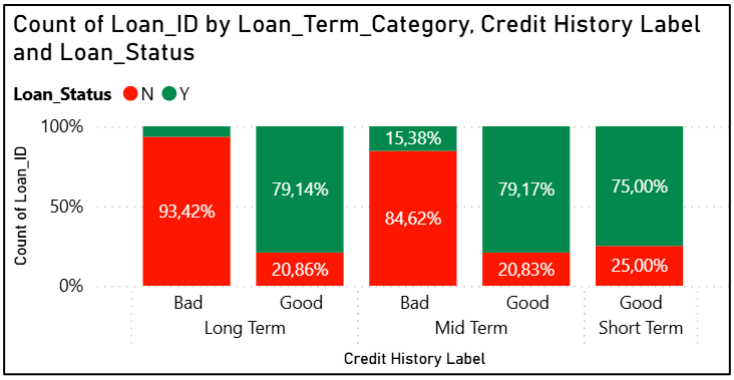


Figure 3. Approved Loans by Loan Term Category and Credit History

On the other hand, Figure 3 shows that credit history is the strongest predictor of loan eligibility. Applicants with good credit history make up almost all approvals across long, medium, and short loan terms. They achieve approval rates between 43 and 50 percent in each loan term category. This pattern highlights that a positive credit history is essential, no matter the loan duration.

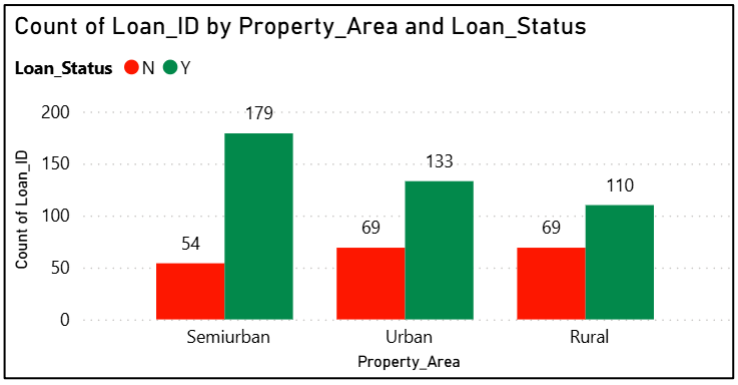


Figure 4. Approved Loans by Property Area

Finally, Figure 4 indicates that the Location-based patterns persist. Applicants from semi-urban areas continue to lead in loan approvals. Urban and rural regions follow. This matches the chi-square hypothesis test results, confirming that property area is a significant factor influencing approvals.

Continuous Variables: *ApplicantIncome, CoapplicantIncome,* and *LoanAmount.*

The Statistical analysis of numerical variables shows high variability in income and loan amounts requested. The average income of applicants is $5,403.45, with a standard deviation of $6,109.04, indicating a dispersed distribution and the possible presence of extreme values, as reflected in the maximum value of $81,000.

Moreover, the co-applicant's income is significantly lower on average ($1,621.24) and has a median of just $1,188.5, suggesting that in many cases there are no additional income contributions (since the first quartile and minimum are $0). As for the loan amount, the mean is $145,752.44 with a high standard deviation of $84,107.44, and values ranging from $9,000 to $700,000. This also suggests normalizing or transforming these variables in the models, especially given their bias toward high values and their wide range.

Table 1. Continues Variables Statistic Analysis

	Applicant Income	**Coapplicant Income**	**Loan Amount**
Mean	$ 5,403.45	$ 1,621.24	$ 145,752.44
Std	$ 6,109.04	$ 2,926.24	$ 84,107.44
Min	$ 150	$ 0	$ 9,000
Q1	$ 2,877.5	$ 0	$ 100,250
Q2	$ 3,812.5	$ 1,188.5	$ 128,000
Q3	$ 5,795	$ 2,297.25	$ 164,750
Max	$ 81,000	$ 41,667	$ 700,000

Considering the above, histograms (Figures 5 & 6) were plotted for each of the continuous variables, and a strong right-skewed trend was identified. Therefore, logarithmic transformations were applied to reduce the impact of outliers, resulting in a considerable improvement in the distribution of each variable and making them suitable for model testing.

However, it was observed that despite the transformation, the variability of co-applicant income continues to be affected by those records where there is no co-applicant as loan support.

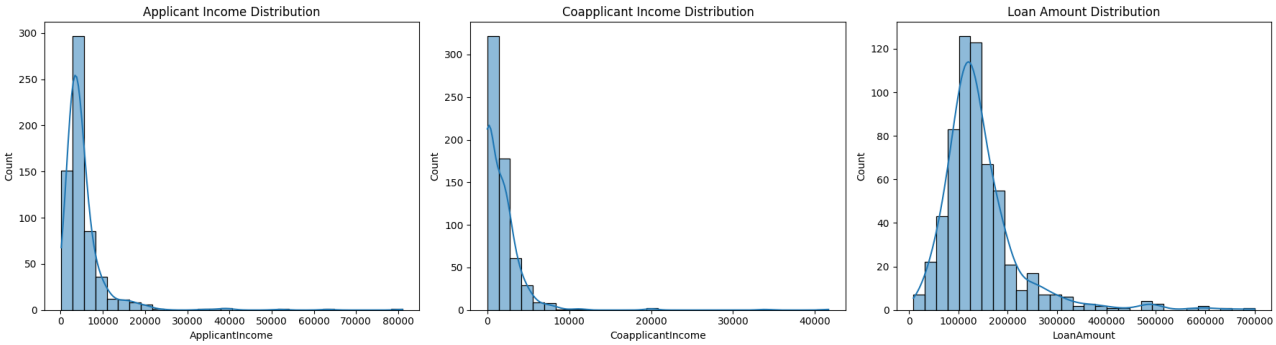


Figure 5. Numerical variables before log transformation

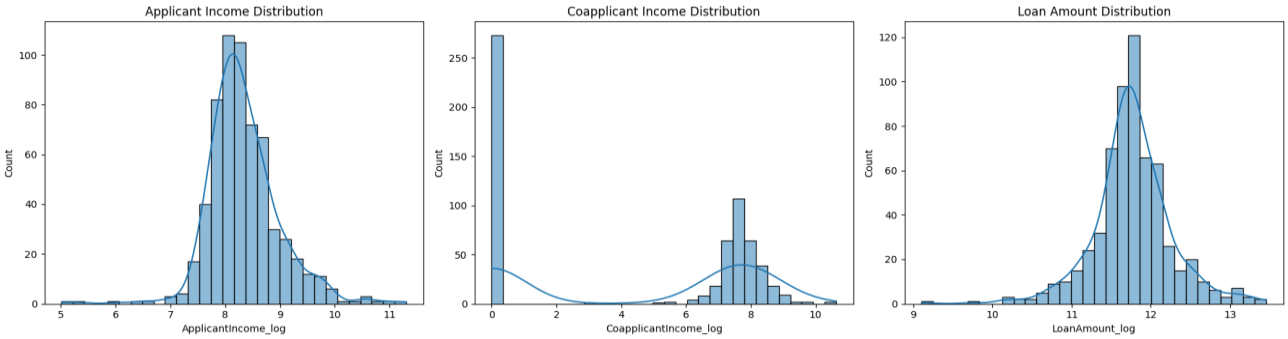


Figure 6. Numerical variables before log transformation

Integrated Insights

By confirming the importance of credit history, education, and property area, the dashboard highlights the smaller role of marital status and dependents. This builds confidence in the key variables that are most likely to affect loan eligibility. The decomposition tree (Figure 7) shows that credit history is the main factor. It breaks down into education, marital status, and property area as other influences. This reinforces the step-by-step reasoning behind loan approval decisions.

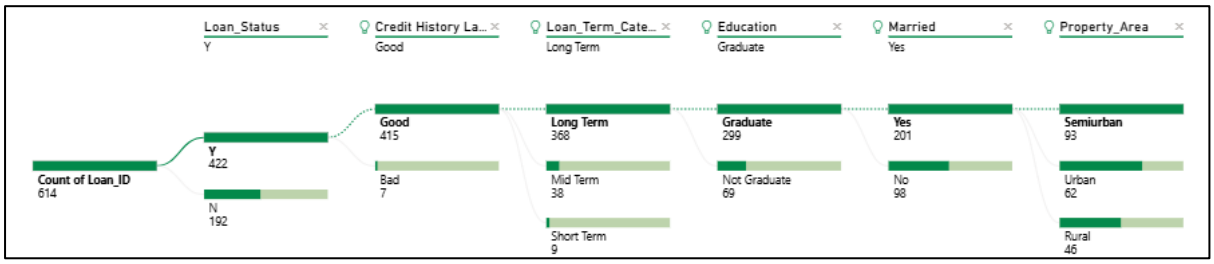


Figure 7. Numerical variables before log transformation

Test and Correlations

To assess if there is a significant association between credit history and loan approval within each loan category, a Chi- Square Test was conducted. The result for Chi-Square Test for long term loan was 157.26 and p-value was 4.47e-36. Since the p-value is less than the significance level (alpha), which is 0.05, we reject the null hypothesis. This indicates that there is a statistically significant association between Credit history and loan status for long term loan category.

The same test was conducted for midterm and short-term loans, and the results obtained were Chi-Square Statistic (15.71), P-value (7.354e-05), and Chi-Square Statistic (0.0), P-value (1.0), respectively Since the p-value for the mid-term loan is less than 0.05 which is the significance alpha, we reject the null hypothesis. This suggests that there is a statistically significant association between Credit history and loan status for this loan term category. Additionally, the p-value for a short-term loan is greater than 0.05. We failed to reject the null hypothesis. This indicates that there is no statistically significant relationship between credit history and loan status for this loan term category.

Moreover, to test whether the educational level (graduate or non-graduate) is associated with loan status, a Chi-square Test was conducted. The results obtained were Chi-square statistics: 4.091 and p-value: 0.043e-08. This indicates that there is a statistically significant correlation between education and loan status; hence, we reject the null hypothesis and accept the alternative hypothesis.

After conducting the Chi-Square Test to assess if there is an association between property area and loan eligibility, the following results were obtained: Chi-Square Statistic- 12.29 and P-value- 0.002.Since the p-value is less than the significance level alpha = 0.05, we reject the null hypothesis and retain alternative hypothesis which indicates that there is a statistically significant association between property area and loan status.

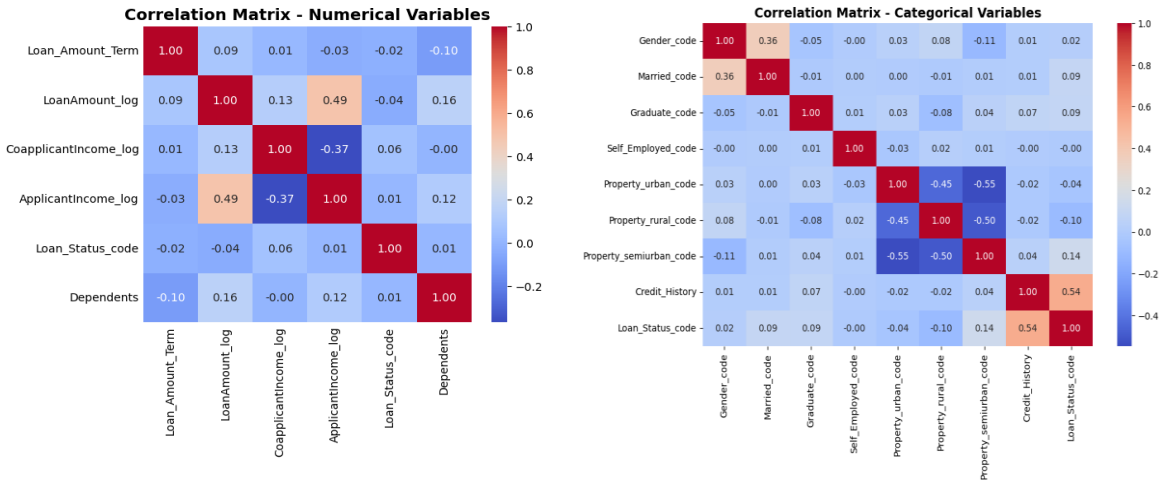


Figure 8. Categorical and Numerical Correlation Matrix (Heatmap)

The correlations (Figure 8) show that credit history (0.54) is the most influential factor in loan approval, while the other variables have very weak relationships. Among the categorical

variables, property location, marital status, and educational level have a slight positive association, while rural properties tend to be slightly more likely to be rejected. Numerical variables, such as income and loan amount, have almost no influence. This suggests that approval depends mainly on credit history, so it would be advisable to include more financial data and increase the sample size to achieve a more reliable model.

Model Building

Three supervised classification models—*Logistic Regression, Random Forest, and XGBoost*—were constructed and contrasted to assess loan approvals. The logistic regression was chosen as the base model due to its interpretive capabilities, which enable analysis of the strength and direction of each variable's influence on the likelihood of approval. In order to identify nonlinear relationships and potential complex interactions between the predictor variables, two ensemble models based on decision trees (*Random Forest and XGBoost*) were subsequently implemented.

Therefore, to determine the most important factors in the loan approval decision, each model was trained and assessed using the same data set. Performance metrics like Accuracy, ROC-AUC Curve, Confusion Matrix, and Classification Report were used, along with an analysis of the relative importance of the variables.

Model Evaluation and Results

Table 2. Logistic Regression, Random Forest, and XGBoost performance comparison

		Presicion	**Recall**	**F1-Score**
Logistic Regression	No (0)	0.93	0.42	0.57
	Yes (1)	0.76	0.98	0.86
Random Forest	No (0)	0.74	0.49	0.59
	Yes (1)	0.77	0.91	0.83
XGBoost Model	No (0)	0.69	0.48	0.56
	Yes (1)	0.76	0.88	0.82

Table 3. Logistic Regression, Random Forest, and XGBoost accuracy comparison

	Acurracy	**Roe Curve**
Logistic Regression	0.78	0.76
Random Forest	0.76	0.76
XGBoost	0.74	0.74

Precision, Recall, and F1-Score

Table 1 presents the precision, recall, and F1-scores for each model across both classes. Logistic Regression achieved very high precision for the negative class (0.93) and high recall

for the positive class (0.98). However, the recall for the negative class was relatively low (0.42), indicating that the model often misclassified negatives as positive.

Random Forest provided a more balanced performance across both classes, with precision values of 0.74-0.77 and recall ranging from o.49(negative class) to 0.91(positive class). Its F1-score for the positive class (0.83) was the highest among the models, showing strong predictive power for class 1 while maintaining reasonable balance.

XGBoost also produced a strong recall for the positive class (0.88) and a relatively high F1-score (0.82). However, its precision was slightly lower compared to Logistic Regression and Random Forest. This indicates that while XGBoost captures positive cases effectively, it is more prone to misclassifying negatives as positives.

On the other hand, the confusion matrices (Figure 9) show similar performance with variations in the classification of approvals and rejections. The detection of approvals was prioritized by logistic regression, which had more false positives (38), but very few false negatives (2). With 33–34 false positives and 11–14 false negatives, Random Forest and XGBoost demonstrated a more consistent balance with a lower overall accuracy.

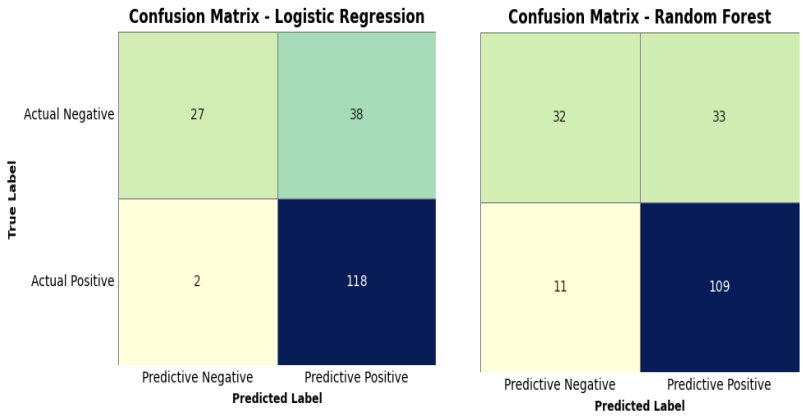


Figure 9. Logistic Regression, Random Forest and confusion matrices comparison

Accuracy and ROC(AUC) Comparison

Table 3 compares the models in terms of overall accuracy and ROC curve (Figure 10). Logistic Regression recorded the highest accuracy (0.78) and ROC value (0.76), followed closely by Random Forest (accuracy =0.76, ROC=0.76) and XGBoost (accuracy = 0.74, ROC = 0.74). These results suggest that all three models perform moderately well in distinguishing between classes, though Logistic Regression slightly outperforms the others in terms of overall classification accuracy.

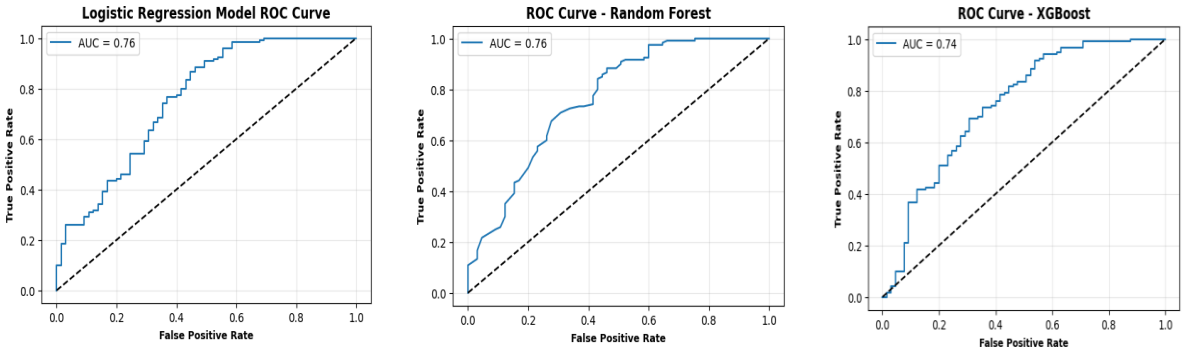


Figure 10. Logistic Regression, Random Forest, and XGBoost ROE curves

Results Interpretation

The evaluation highlights important trade-offs between the models. Logistic Regression is the most accurate overall and provides the best ROC value, however it is biased toward predicting positives, leading to weaker performance on negatives. Random Forest strikes the best balance across metrics, achieving the strongest F1- score for the positive class (0.83), making it well suitable for contexts where recall and balanced performance are critical. XGBoost performs competitively with high recall for positives, though at the cost of lower precision and slightly reduced overall accuracy.

In the logistic regression model, the estimated coefficients indicate the log-odds effect of each variable on the probability of loan approval. Figure 11 shows that Credit History (3.21) is by far the most significant predictor, according to the results. While Graduate (0.22) provides a marginal, other variables like Married (0.62) and Property semiurban (0.58) also have moderately positive effects, Gender code (-0.16), Loan Amount (-0.20), Property urban (-0.25), and Property rural (-0.33), on the other hand, have detrimental effects and lower the likelihood of approval. These findings support the idea that credit history is the primary determinant, with the other factors having a smaller impact on credit risk.

It is important to mention that loan approval has a negative coefficient associated with it, meaning that the larger the loan, the less likely it is to be approved.

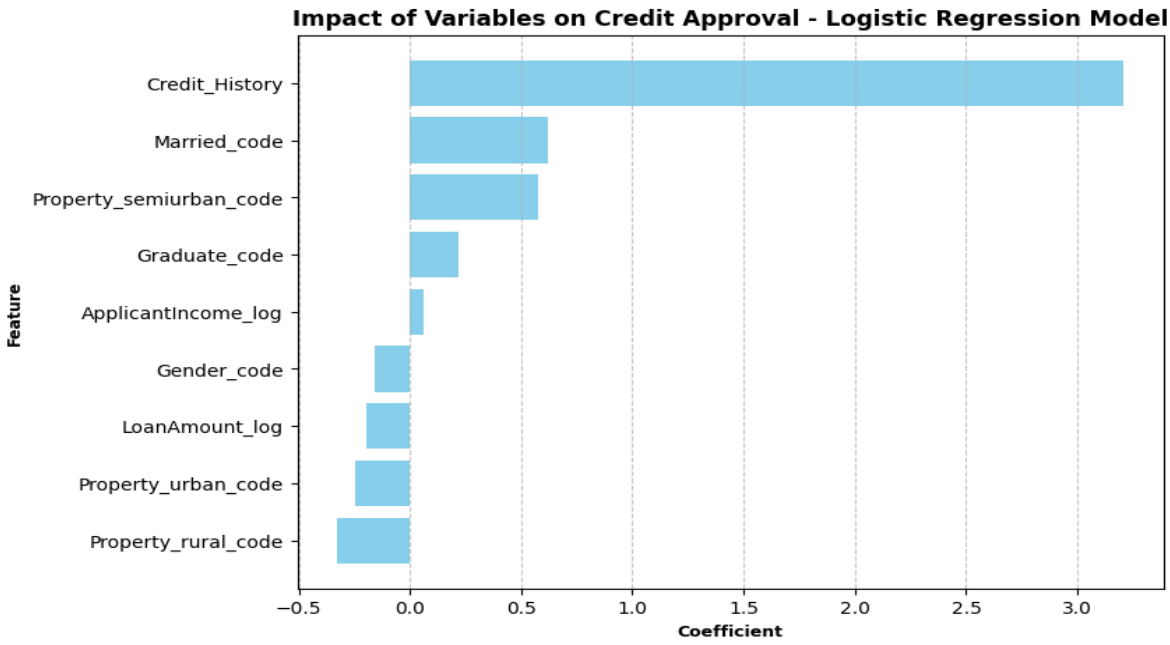


Figure 11. Logistic Regression Model Impact Analysis

The importance of variables for Random Forest (Figure 12) shows that Credit_History (0.253) is still the most relevant predictor, followed by Applicant Income (0.221) and Loan Amount (0.206). This suggests that the amount requested, repayment capacity, and credit history are all important classification factors. The model prioritizes financial factors over personal attributes, as evidenced by the moderate effects of variables like Dependents (0.052) and Coapplicant Income (0.123).

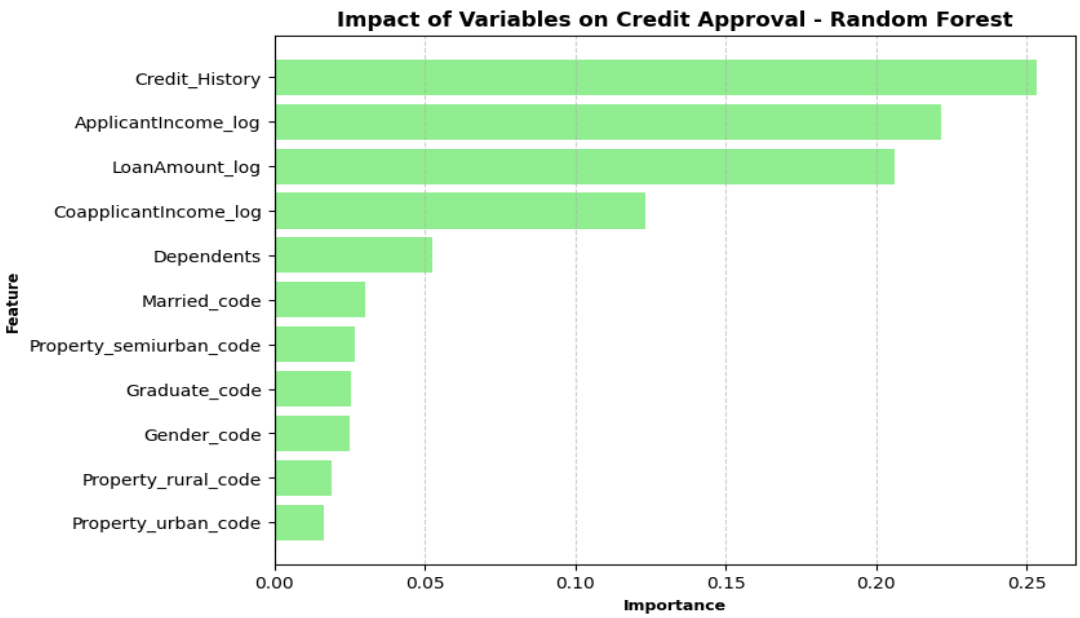


Figure 12. Random Forest Model Impact Analysis

According to Figure 13, Credit_History (0.529) is once again the most influential variable in XGBoost, showing its crucial role in loan approval. While financial variables such

as Loan Amount, Applicant Income, and Coapplicant Income have comparable importance (~0.048), Property semiurban (0.092) appears to be the second most relevant factor. Indicating that XGBoost prioritizes financial information and credit history while also capturing some influence of property location on the probability of approval, the remaining variables—marital status, gender, and property location—contribute slightly.

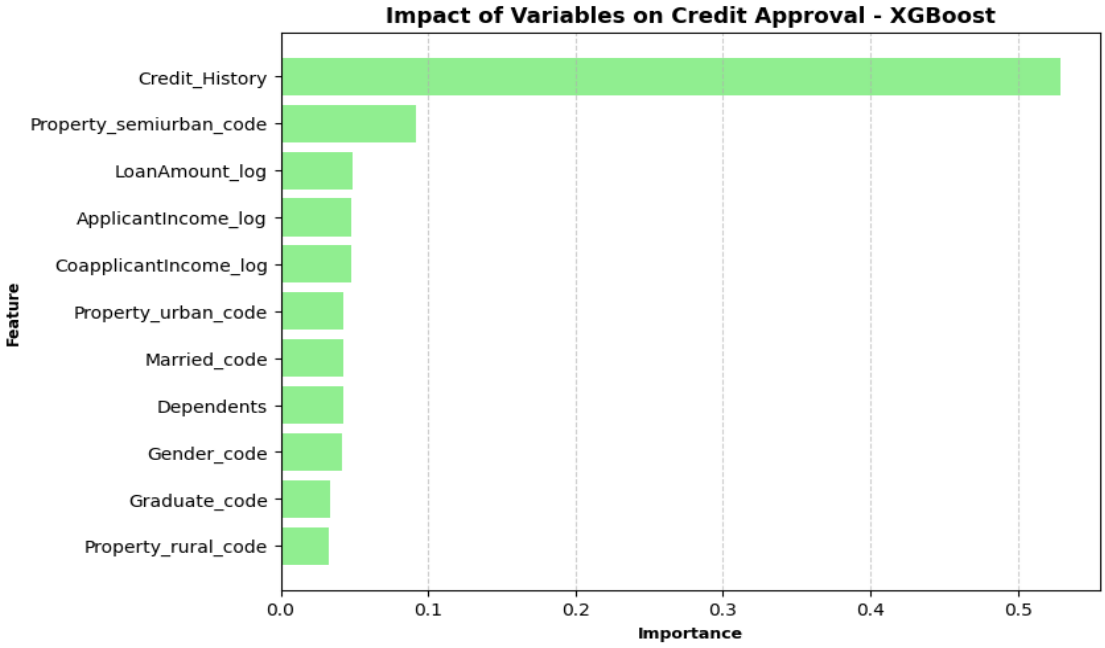


Figure 13. XGBoost Model Impact Analysis

Conclusion and Recomendations

A positive credit history is the deciding factor in loan approval, as proven by the fact that Credit History consistently remains the most influential variable in logistic regression, Random Forest, and XGBoost when comparing the three models created to analyze loan approval. While Random Forest and XGBoost place a greater importance on financial factors like the applicant's and co-applicant's income and the amount requested, logistic regression also emphasizes the significance of demographic variables like marital status, education, and property location.

For long- and medium-term loans, the Chi-Square Test results statistically support the significant correlation between credit history and loan status; however, no significant correlation is found for short-term loans, suggesting that approval in the latter case depends on other factors or discretionary decisions by the bank.

Overall, the models offer solid information for the profiles that increase the likelihood of approval: a customer who has a good credit history, makes enough money to cover the requested amount, is married, ideally a graduate, and lives in a semi-urban area, has the best chance of getting approved. This picture illustrates how the team can comprehend the primary

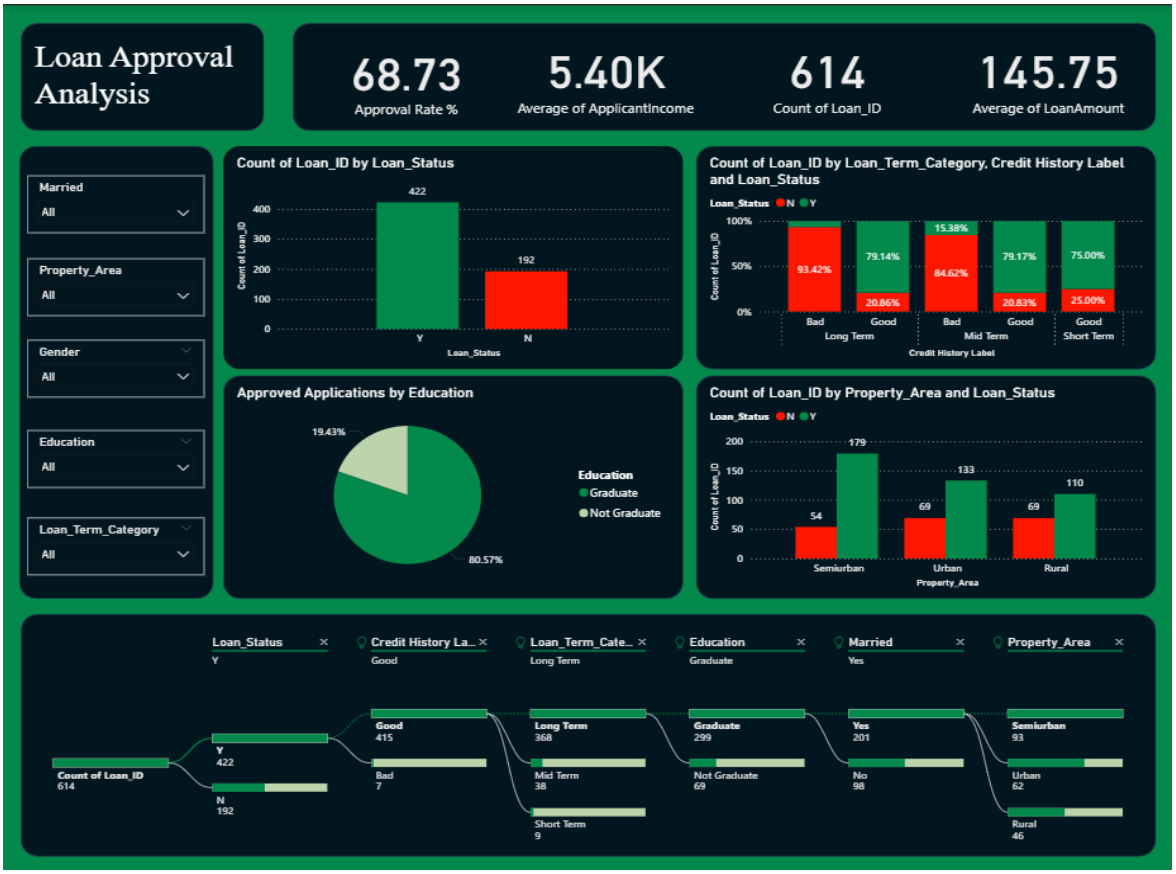
factors that determine loan approval by combining demographic and financial data to inform credit decisions.

Even though the models produced have acceptable accuracy levels, their predictive power could be improved by significantly expanding the dataset, which would enable improved generalization and bias reduction. Similarly, the addition of more variables involving debt-to-income ratio, job stability, and financial history would be useful to create more robust and accurate models for loan approval prediction.

References

Eligibility Prediction for Loan. https://www.kaggle.com/datasets/devzohaib/eligibility-prediction-for-loan.

Apendix


Loan Approval Analysis
68.73
Approval Rate %
5.40K
Average of ApplicantIncome
614
Count of Loan_ID
145.75
Average of LoanAmount
Married
All
Property_Area
All
Gender
All
Education
All
Loan_Term_Category
All
Count of Loan_ID by Loan_Status
422
192
Count of Loan_ID
Loan_Status
Count of Loan_ID by Loan_Term_Category, Credit History Label and Loan_Status
Loan_Status N Y
93.42%
79.14%
20.86%
15.38%
84.62%
79.17%
20.83%
75.00%
25.00%
Bad
Good
Long Term
Mid Term
Short Term
Credit History Label
Approved Applications by Education
19.43%
80.57%
Education
Graduate
Not Graduate
Count of Loan_ID by Property_Area and Loan_Status
54
179
69
133
69
110
Semiurban
Urban
Rural
Property_Area
Loan_Status
Y
Credit History La...
Good
Loan_Term_Cate...
Long Term
Education
Graduate
Married
Yes
Property_Area
Count of Loan_ID
614
422
192
415
7
368
38
9
299
69
201
98
93
62
46